## INDUS UNIVERSITY

### Monthly Progress Report - Software Group Project

Academic Year: 2026-27  Month: July

Project Title: House price prediction using AI/ML

Project Domain: AI/ML

Group Number: 50  Project Guide: Dhwani Mam

## Student Details:

| Enrolment No. | Student Name | Contribution | Signature |
|---|---|---|---|
| IU2341050103 | Ravi Patel | ➢ Project Introduction<br>➢ Dataset<br>➢ Dataset Understanding<br>➢ Missing values | |
| IU2341050105 | Kenil Patel | ➢ Data Cleaning<br>➢ Feature Creation (BHK)<br>➢ total_qrft conversion<br>➢ Duplicate Removal | |
| IU2341050114 | Vishnu Jikadara | ➢ Outlier Analysis<br>➢ Final Cleaned Dataset<br>➢ Summary<br>➢ EDA (Exploratory Data Analysis) | |

## Project Objective:

To develop a House Price Prediction System using Machine Learning. The project aims to clean and analyze the Bengaluru House Price dataset, prepare high-quality data, and build an accurate model for predicting house prices based on different property features.

## Monthly Work Completed

- Project title finalized.
- Bengaluru House Price dataset collected.
- Dataset understanding completed.
- Missing values handled.
- Duplicate records removed.
- Data cleaning performed.
- size column converted into `bhk`.
- `total_sqft` column cleaned and converted to numeric values.
- Created `price_per_sqft` feature.
- Outlier analysis performed.
- Outliers removed using domain knowledge.
- Final cleaned dataset prepared.
- Basic EDA completed.
- Distribution analysis performed using histograms and countplots.
- Relationship analysis performed using scatter plots and boxplots.
- Correlation matrix and heatmap generated for numerical features.

## Challenges Faced

- The dataset contained missing values.
- The `total_sqft` column had inconsistent formats.
- Outliers affected the data distribution.
- Some categorical values required preprocessing.

## Next Month Plan

- Perform feature engineering.
- Apply One-Hot Encoding.
- Split data into training and testing sets.

- Train multiple Machine Learning models.
- Compare model performance.
- Select and save the best model.

## Guide Remarks

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Overall Progress : 20 %

Student Signatures:

- Ravi Patel : _____
- Kenil Patel : _____
- Vishnu Jikadara: _____

Project Guide: _____